

Javascript:

// Online Java Compiler

// Use this editor to write, compile and run your Java code online

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
class Student {
```

```
    private String id;
```

```
    private String name;
```

```
    private ArrayList<Double> grades;
```

```
    public Student(String id, String name) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        grades = new ArrayList<>();
```

```
    }
```

```
    public String getId() {
```

```
        return id;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
    }
```

```
    public ArrayList<Double> getGrades() {
```

```
        return grades;
```

```
    }
```

```
public void addGrade(double grade) {  
    grades.add(grade);  
}
```

```
public boolean hasGrades() {  
    return !grades.isEmpty();  
}
```

```
public double getAverage() {  
    if (grades.isEmpty()) return 0;  
  
    double sum = 0;  
    for (double g : grades)  
        sum += g;  
  
    return sum / grades.size();  
}
```

```
public double getHighest() {  
    if (grades.isEmpty()) return 0;  
  
    double max = grades.get(0);  
  
    for (double g : grades)  
        if (g > max)  
            max = g;  
  
    return max;  
}
```

```
public double getLowest() {  
    if (grades.isEmpty()) return 0;  
  
    double min = grades.get(0);  
  
    for (double g : grades)  
        if (g < min)  
            min = g;  
  
    return min;  
}
```

```
public String getLetterGrade() {  
    double avg = getAverage();  
  
    if (avg >= 90) return "A";  
    else if (avg >= 80) return "B";  
    else if (avg >= 70) return "C";  
    else if (avg >= 60) return "D";  
    else return "F";  
}  
}
```

```
public class Main {  
  
    static ArrayList<Student> students = new ArrayList<>();  
    static Scanner sc = new Scanner(System.in);  
  
    public static void main(String[] args) {
```

```
while (true) {

    System.out.println("\n===== STUDENT GRADE TRACKER =====");

    System.out.println("1. Add Student");

    System.out.println("2. Add Grades");

    System.out.println("3. View Student Details");

    System.out.println("4. Summary Report");

    System.out.println("5. Remove Student");

    System.out.println("6. Exit");

    System.out.print("Enter Choice: ");

    int choice = sc.nextInt();

    sc.nextLine();

    switch (choice) {

        case 1:

            addStudent();

            break;

        case 2:

            addGrades();

            break;

        case 3:

            viewStudent();

            break;

        case 4:

            summaryReport();
```

```
break;
```

```
case 5:
```

```
    removeStudent();
```

```
    break;
```

```
case 6:
```

```
    System.out.println("Thank You!");
```

```
    System.exit(0);
```

```
default:
```

```
    System.out.println("Invalid Choice!");
```

```
}
```

```
}
```

```
}
```

```
static void addStudent() {
```

```
    System.out.print("Enter Student ID: ");
```

```
    String id = sc.nextLine();
```

```
    System.out.print("Enter Student Name: ");
```

```
    String name = sc.nextLine();
```

```
    for (Student s : students) {
```

```
        if (s.getId().equals(id)) {
```

```
            System.out.println("Student ID already exists!");
```

```
            return;
```

```
        }
```

```
    }
```

```
students.add(new Student(id, name));  
  
System.out.println("Student Added Successfully!");  
  
}
```

```
static void addGrades() {
```

```
    System.out.print("Enter Student ID: ");
```

```
    String id = sc.nextLine();
```

```
    Student student = findStudent(id);
```

```
    if (student == null) {
```

```
        System.out.println("Student Not Found!");
```

```
        return;
```

```
    }
```

```
    System.out.print("How many grades? ");
```

```
    int n = sc.nextInt();
```

```
    for (int i = 1; i <= n; i++) {
```

```
        System.out.print("Enter Grade " + i + ": ");
```

```
        double grade = sc.nextDouble();
```

```
        student.addGrade(grade);
```

```
    }
```

```
    sc.nextLine();
```

```
    System.out.println("Grades Added Successfully!");
```

```
}
```

```
static void viewStudent() {

    System.out.print("Enter Student ID: ");

    String id = sc.nextLine();

    Student s = findStudent(id);

    if (s == null) {

        System.out.println("Student Not Found!");

        return;

    }

    System.out.println("\n----- STUDENT DETAILS -----");

    System.out.println("ID: " + s.getId());

    System.out.println("Name: " + s.getName());

    System.out.println("Grades: " + s.getGrades());

    if (s.hasGrades()) {

        System.out.printf("Average: %.2f\n", s.getAverage());

        System.out.printf("Highest: %.2f\n", s.getHighest());

        System.out.printf("Lowest: %.2f\n", s.getLowest());

        System.out.println("Letter Grade: " + s.getLetterGrade());

    }

}

static void summaryReport() {

    if (students.isEmpty()) {

        System.out.println("No Students Available!");

    }

}
```

```
    return;
```

```
}
```

```
System.out.println("\n===== SUMMARY REPORT =====");
```

```
for (Student s : students) {
```

```
    System.out.println("-----");
```

```
    System.out.println("ID: " + s.getId());
```

```
    System.out.println("Name: " + s.getName());
```

```
    if (s.hasGrades()) {
```

```
        System.out.printf("Average: %.2f\n", s.getAverage());
```

```
        System.out.printf("Highest: %.2f\n", s.getHighest());
```

```
        System.out.printf("Lowest: %.2f\n", s.getLowest());
```

```
        System.out.println("Grade: " + s.getLetterGrade());
```

```
    } else {
```

```
        System.out.println("No Grades Added");
```

```
    }
```

```
}
```

```
}
```

```
static void removeStudent() {
```

```
    System.out.print("Enter Student ID: ");
```

```
    String id = sc.nextLine();
```

```
    Student s = findStudent(id);
```

```
    if (s == null) {
```



```
        System.out.println("Student Not Found!");  
        return;  
    }
```

```
        students.remove(s);  
        System.out.println("Student Removed Successfully!");  
    }
```

```
static Student findStudent(String id) {  
  
    for (Student s : students) {  
        if (s.getId().equals(id))  
            return s;  
    }  
  
    return null;  
}  
}
```

Output

```
===== STUDENT GRADE TRACKER =====
```

1. Add Student
2. Add Grades
3. View Student Details
4. Summary Report
5. Remove Student
6. Exit

```
Enter Choice: 2
```

```
Enter Student ID: 2
```

```
Student Not Found!
```

```
===== STUDENT GRADE TRACKER =====
```

1. Add Student
2. Add Grades
3. View Student Details
4. Summary Report
5. Remove Student
6. Exit

```
Enter Choice:
```